Determination: the canonical model validates exactly the theorems of Sigma

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the Truth Lemma is referenced by its English name only in the source text (no olref) - prereq pins it by section position; the surrounding world-set notation W^Sigma is kept.

Editorial note. The label thm:determination names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Truth Lemma. Same part, the Truth Lemma proposition (source line 1379 region) - cited in prose, not by olref macro. Lines 649-651 and 657-658: the two conversions between global truth and set membership at each world.
- Provability as membership in every maximal extension. Same part, \ollabel{cor:provability-characterization} (source line 1049, own card). Line 651: at Gamma empty, membership in all complete consistent sets becomes Sigma-provability.
- Deductive closure of complete consistent sets. Same part, \ollabel{prop:ccs-properties} item (1) (line 504 region). Line 655: Sigma proving A pushes A into every maximal Delta.

Source reference. Original proposition prop-ccs-closed: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 522–522. Prerequisite pointer invoked at proof line(s) 655; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-properties: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 519–519. Prerequisite pointer invoked at proof line(s) 655; it names a standard fact of the

same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original corollary cor-provability-characterization: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1093–1093. Prerequisite pointer invoked at proof line(s) 651; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Determination)**.** $M^\Sigma \models A[]$ *if and only if* $\Sigma \vdash A$.

Proof. If $M^\Sigma \models A[]$, then for every complete Σ-consistent Δ, we have $M^\Sigma \models A[]\Delta]$. Hence, by the Truth Lemma, $A \in \Delta$ for every complete Σ-consistent Δ, whence by cor-provability-characterization (with $\Gamma = \emptyset$), $\Sigma \vdash A$.

Conversely, if $\Sigma \vdash A$ then by prop-ccs-propertiesprop-ccs-closed, every complete Σ-consistent Δ contains A, and hence by the Truth Lemma, $M^\Sigma \models A[]\Delta]$ for every $\Delta \in W^\Sigma$, i.e., $M^\Sigma \models A[]$. □